Investor Behavior Across Different Interest Rate Regimes

Work In Progress

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Abstract

In this paper, I measure the loss aversion of simulated investors with respect to portfolio returns across different interest rate regimes. Three investor profiles, risk-averse, risk-neutral, and risk-seeking, are modeled under a Cumulative Prospect Theory (CPT) framework, and loss aversion is numerically identified by minimizing the squared difference between the CPT-optimal and Mean-Variance efficient portfolio allocations. The findings demonstrate that implied loss aversion is significantly dependent on the prevailing interest rate environment. Specifically, loss aversion decreases in low-interest-rate regimes, encouraging a statistically and economically significant shift toward riskier assets. Conversely, in high-interest-rate regimes, loss aversion increases, leading to a portfolio re-allocation towards safer assets. These results provide strong evidence that the market's psychological state, as captured by loss aversion, responds systematically to macroeconomic uncertainty and the prevailing stance of monetary policy.

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1 Introduction

Observations of investor behavior often diverge from prevailing models in financial economics, which tend to build on expected utility theory (EUT) and its leading assumption of rational decision-making. While EUT has long been a cornerstone in economic models of asset allocation, it does not account for the empirical discrepancies between its predictions of asset allocation and observed investor behavior. A more accurate depiction of real-world decision-making, particularly in the face of uncertainty and risk, is offered by Cumulative Prospect Theory (CPT), a framework developed by Tversky and Kahneman (1992). CPT formalizes the earlier proposed theory, prospect theory (see Kahneman and Tversky (1979)), which advances a theory of utility from monetary gambles that emphasizes decision making relative to a reference point. Subjective utility is derived from the gains and losses realized with respect to an individual's reference point. Individuals facing prospective gambles under uncertainty are subject to certain psychological effects that lead to a greater disutility of losses relative to the utility of gains, a phenomenon captured by the model's loss aversion parameter. Rather than using a non-linear, monotonic transformation of individual probabilities associated with outcomes of a gamble, as in the first iteration of prospect theory, CPT integrates the cumulative form to transform the entire probability distribution. Decision weights, in place of physical probabilities, are defined to maintain prospect theory's non-linearity over gains and losses, or concave (risk-averse) over gains and convex (risk-seeking) over losses. This notable addition of the cumulative form also enables the consideration of gambles with any number of outcomes, and maintains stochastic dominance in gambles involving uncertainty. Results of lab experiments in Tversky and Kahneman (1992) show overweighting of small probabilities and underweighting of large probabilities, in addition to risk aversion for gains and risk seeking for losses. These features of CPT offer key insights into how investors assess risk and uncertainty in making asset allocation decisions.

Behavioral finance draws on the non-trivial results of works such as CPT to enhance the understanding of financial decision-making, and it has significantly benefited from the loss aversion framework. One major application is in understanding the equity premium puzzle—the long-standing observation that equities have historically provided higher returns than bonds, despite the higher risk premium observed in the data and in theoretical models,

such as consumption-based asset pricing. This puzzle has been addressed by Benartzi and Thaler (1995) through their concept of myopic loss aversion, whereby the combination of frequent portfolio evaluations and heightened sensitivity to short-term fluctuations leads investors to increase their aversion to losses and demand higher returns from equities.

While much of the previous literature has focused on problems such as the equity premium puzzle or the differences in investing tendencies between professional and retail investors, this paper focuses on whether investors' loss aversion changes over time and under distinct interest rate regimes. As the U.S.'s central bank, the Federal Reserve influences short-term market interest rates via the Federal Funds Rate to effectively implement its dual mandate of price stability and maximum employment. Directional changes in short-term interest rates tend to impact the prices of different financial assets, giving way to a flurry of trading across different financial markets. Thus, changes in the Federal Reserve's monetary policy can lead to a period of increased volatility in asset prices at the time of such policy implementation, in turn affecting investor behavior. Policy implementations are associated with evovling macroeconomic conditions and changes in the expectations of future economic variables, which can last during the regime set forth by a policy announcement. By simulating portfolios representing risk-averse, risk-neutral, and risk-seeking investors, this paper examines how loss aversion of different investors varies over time and under different interest rate environments. A key aim of this analysis is to measure a loss aversion parameter that is time-varying, similar in spirit to other recent works (see Liu et al. (2025)). Extracting estimates of this parameter over different sequential interest rate regimes would yield important results for the latent behavioral aspects of investors in different interest rate environments.

The paper continues as follows. Section 2 discusses the relevance of such an examination to the current state of U.S. financial markets, with a subsequent review of the application of prospect theory to the measurement of investor loss aversion under different interest rate regimes. Descriptive statistics are provided to help motivate the analysis and illustrate the historical relationship between financial asset returns and the Effective Federal Funds Rate (federal funds rate). Section 3 describes the data used in the analysis, and Section 4 describes the construction of representative portfolios for the three selected investor risk profiles, as well as the distinction of interest rate regimes. Section 5 provides a subsequent analysis of each portfolio simulated during historical interest rate regimes, and Section 6 encapsulates

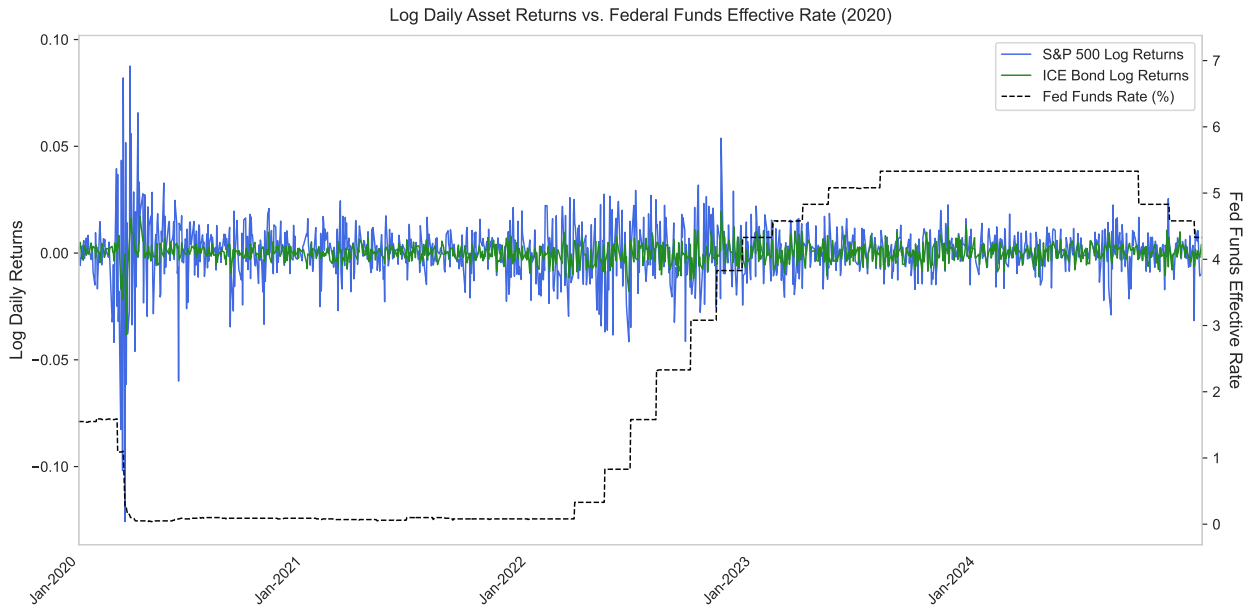
the findings of the analysis. Finally, Section 7 provides concluding remarks.

2 Prospect Theory, Investor Behavior, and Monetary Policy

2.1 Importance of Loss Aversion in Different Interest Rate Environments

Financial market turbulence can often be a result of central bank policy implementations, as is particularly observed during the global COVID-19 pandemic up to the end of 2024. In the year leading up to the pandemic, interest rates began rising after being kept at low levels, a policy decision that supported economic recovery after the 2008 Financial Crisis. Following the onset of the global pandemic in March of 2020, the Federal Open Market Committee (FOMC) lowered its target range for the federal funds rate to near-zero levels to help alleviate the pandemic's shock to economic activity. As global economies recuperated from the effects of the pandemic, PCE inflation began to rise leading the FOMC to implement rate increases in order to slow the acceleration in prices. In both of these instances, a decrease and an increase in the federal funds rate, we see a noticeable spike in financial market activity, graphically depicted below.

Figure 1: Log Asset Returns and the Effective Federal Funds Rate, 2020 to 2024



Volatility in financial asset prices can, in part, be due to adjacent factors unrelated to the implementation of monetary policy itself, namely factors associated with the real economy, asset fundamentals, or "animal spirits." Changes in real economic activity can both precede and follow monetary policy implementations, an area extensively addressed in the literature and reviewed below. Volatility preceding policy is based on *ex ante* information, as the directionality of such an implementation is still presumably outside of the information set. Immediately after policy announcements, heightened volatility persists as market participants incorporate *ex post* information, with the magnitude of market volatility often depending on whether the policy is a surprise (*ex ante* expectations do not align with *ex post* observations) or aligns with market expectations. The exact channel through which monetary policy impacts asset prices varies substantially for different asset classes. However, when sparked by changes in short-term interest rates, market volatility invariably leads to significant re-allocations of assets across many portfolios.

Short-term fluctuations in asset prices are often driven by the trading behavior of various investors due to exogenous and endogenous factors, such as macroeconomic sentiment

and herding behavior, respectively. Two broad investor types are typically identified: institutional and individual (or retail). Institutional investors presumably have well-developed tools and professional experience to aid in portfolio adjustments when volatility spikes. Retail investors, however, make asset allocation decisions without such resources. When under heightened economic uncertainty, a choice problem already made difficult by the lack of perfect foresight becomes even more challenging. With a focus on the retail investor, behavioral finance provides a useful framework for capturing the psychological effects of economic activity on investing behavior. One important result of behavioral finance is the impact of framing, or how the presentation of a prospective gamble influences an individual's subjective utility, as highlighted by Tversky and Kahneman (1992). The framing process that individuals experience during periods of market volatility and the months following them dramatically changes how they perceive their investment decisions. This is accentuated by the notion of loss aversion.

2.2 Evolution of Prospect Theory and Behavioral Finance

Cumulative Prospect Theory (CPT) has decidedly become an alternative approach to describing the preferences and behaviors of individuals. Kahneman and Tversky (1979) initially developed Prospect Theory, arguing for the use of prospective utility in place of expected utility for modeling an agent's decision-making process concerning various gambles under uncertainty. Instead of the typical utility function weighted by probabilities of events, they introduce prospective utility as the product of a value function and decision weights. The value function considers changes in wealth relative to a reference point instead of totalities, leading to the core idea of gains and losses. Decision weights replace event probabilities to exhibit non-linearity in the probability distribution associated with the utility derived from these gains and losses. The transformation of probabilities emphasizes an individual's tendency to overestimate low probability events (i.e., assign a higher probability weight) and underestimate the reverse. Together, the value function and decision weights result in the fundamental observation of loss aversion, or a greater sensitivity towards losses than gains.

CPT builds upon this most notably with the cumulative form for constructing decision weights, as well as empirical results supporting loss aversion. Tversky and Kahneman (1992)

find that how different gambles are presented to an agent influences their evaluation of such gambles, denoting this as a framing process. They provide a formal definition of decision weights and their preferred non-linear probability function. Similar to the earlier work on prospect theory, CPT's value function is a piecewise power function that is positive and concave for gains and negative and convex for losses, with the slope of losses being larger than that of gains. Through lab experiments with individual participants, they show overweighting of small probabilities and underweighting of large probabilities, supporting their hypothesis of risk aversion for gains (concave and flat) and risk seeking for losses (convex and steep).

Among the many fields of economics that have applied these advancements in prospect theory, behavioral finance has reaped a significant benefit given the focus on monetary gambles under risk and uncertainty. Mehra and Prescott (1985) provide an explicit depiction of how the returns on equities have historically exceeded the returns on bonds. With a higher average return over the long run, it would seem that equities are more optimal to hold in an investor's portfolio. Despite this, many tend to hold a mix of equities and bonds, or, for those more averse to the sporadic movement of equity prices, all bonds, justifying the decision with the safe and reliable guarantee of a fixed-income security. Benartzi and Thaler (1995) add a behavioral perspective to this long-standing quandary in financial economics. They argue for myopic loss aversion of investors, or the tendency to frequently evaluate portfolios and observe large short-term fluctuations in equities. Myopic loss aversion instills the perception that stocks carry a higher degree of risk, and investors, therefore, demand a higher return relative to bonds, offering a behavioral explanation for the equity premium puzzle.

Extending this analysis to institutional investors and university endowments, Benartzi and Thaler find similar results. Further analysis of the discrepancies between different market participants has been conducted. Regarding non-institutional investors, Haigh and List (2005) and Abdellaoui, Bleichrodt, and Kammoun (2013) separate the loss aversion parameters of individual and professional investors—such as traders—finding the presence of a higher degree of loss aversion for the latter group of investors. Notwithstanding these notable findings, I focus my analysis of loss aversion on that of the individual investor for reasons discussed below.

This combination of economics and psychology allows for another perspective from which to understand the complexities of financial markets. Economists have long attempted to cap-

ture the mechanics of these markets in theory, but have met significant difficulty in obtaining consistent empirical validation. Goetzmann (2000) remarks on the dynamic nature of markets and the attempts to quantify their nature, yet offers caution in making use of such tools. Loss aversion utilizes an alternative framework relative to traditional financial economics models. Yet, to meet halfway with the warnings of Goetzmann and others, even such a robust framework of human behavior may not capture the entire picture. It does, nonetheless, offer a strong perspective from which to draw key insights.

2.3 Monetary Policy as a Framework for Investor Behavior Under Prospect Theory

Loss aversion in the context of financial markets is found by Olsen (1998) to generate larger negative price movements relative to positive price movements of financial assets. This reflection of human behavior—captured by prospect theory's gains and losses—in financial markets gives rise to the relevance of a behavioral approach, as well as the importance of measuring investor loss aversion and how it varies across time as macroeconomic and financial market conditions change.

Monetary policy often comes with a changing macroeconomic environment, and policy implementations play a role in the path of asset prices, typically pushing investors to gravitate towards certain asset classes. An example of such a course of events can be illustrated with two assets, bonds and stocks, as follows. The FOMC implements target ranges for the federal funds rate—the interest rate borne by depository institutions for borrowing funds in overnight markets. A ripple effect ensues in the spread between the federal funds rate and short-term market interest rates as these institutions adjust the rates they offer on their lending products. Existing fixed-income products, such as government bonds, change in pricing as newly issued products carry higher or lower rates, depending on the policy decision, increasing the attractiveness of newly issued fixed-income products. This gives rise to the well-known inverse relationship between bond prices and interest rates. Changes in the yields of short-term and longer-term bonds tend to affect the fundamental valuation of cash-flow-dependent public companies, such as those in the technology sector. A policy announcement also carries the implication of evolving economic conditions, which may boost or dampen the prospects

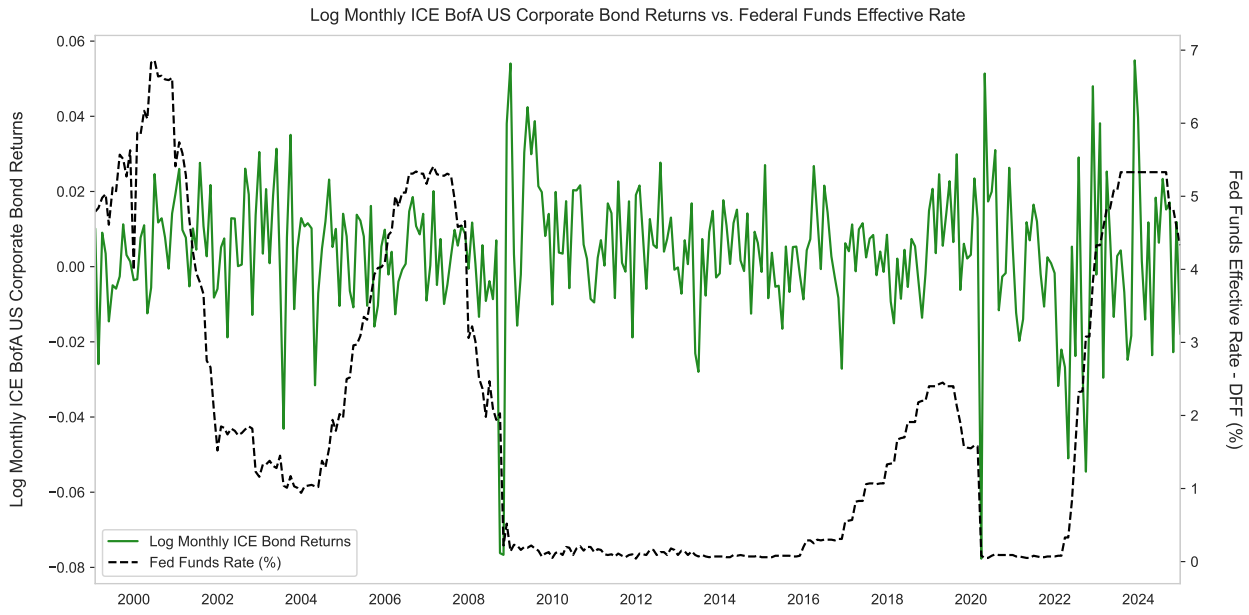
of aggregate consumption and the perception of expected future corporate profits, leading to changes in the prices of non-cash-flow dependent public equities. Following this intuition, monetary policy implementations would have a significant impact on both classes of securities through the changes in short-term market interest rates and the addition of news—not known up until the announcement—into the information set of investors.

Though a large body of empirical literature examining the effects of asset prices on monetary policy exists, papers addressing the opposite—monetary policy's effect on asset prices—have notably increased over the past two decades. This pool of research offers strong empirical evidence for policy decisions to influence the volatility of financial securities, both directly and indirectly, through changes in short-term interest rates via open-market operations and by changing the future expectations of investors regarding economic growth by way of policy announcements.

In analyzing the effect of monetary policy on the volatility of stocks, Bomfim (2000) finds that volatility responds to announcements made by the FOMC on policy decision dates, and the magnitude of the volatility is heightened in the presence of unanticipated policy decisions. Rigobon and Sack (2004) support and expand upon this using both an instrumental variables and generalized-method-of-moments approach. They compare their findings to the ordinary least squares—labeled "event-study" approaches—previously employed by Bomfim (2000) and others. By estimating the effect of a change in the volatility of monetary policy on the (relatively) constant volatility of asset prices, Rigobon and Sack complement the prior work of others, such as Bomfim (2000), in empirically showing that monetary policy has a large effect on asset prices. Specifically, Rigobon and Sack find that an unexpected increase in the federal funds rate is associated with a fall in the prices of both equities and bonds. For equities, this seems to support the present valuation of stocks. They extend the analysis to bonds with maturities of six or more months and find a smaller effect for longer-term bonds than shorter-term bonds.

To illustrate this, Figures 2 and 3 show monthly time series data on the logged returns of the ICE Bank of America Corporate Bond Index and the Standard and Poors (S&P) 500 Index from 1999 to 2024. Both equities and bonds exhibit large swings in logged returns surrounding changes to the federal funds rate, with equities showing a wider spread than bonds. Policy announcements and implementations seem to be immediately followed by changes in

Figure 2: Log Corporate Bond Returns and the Federal Funds Effective Rate, 1999 to 2024

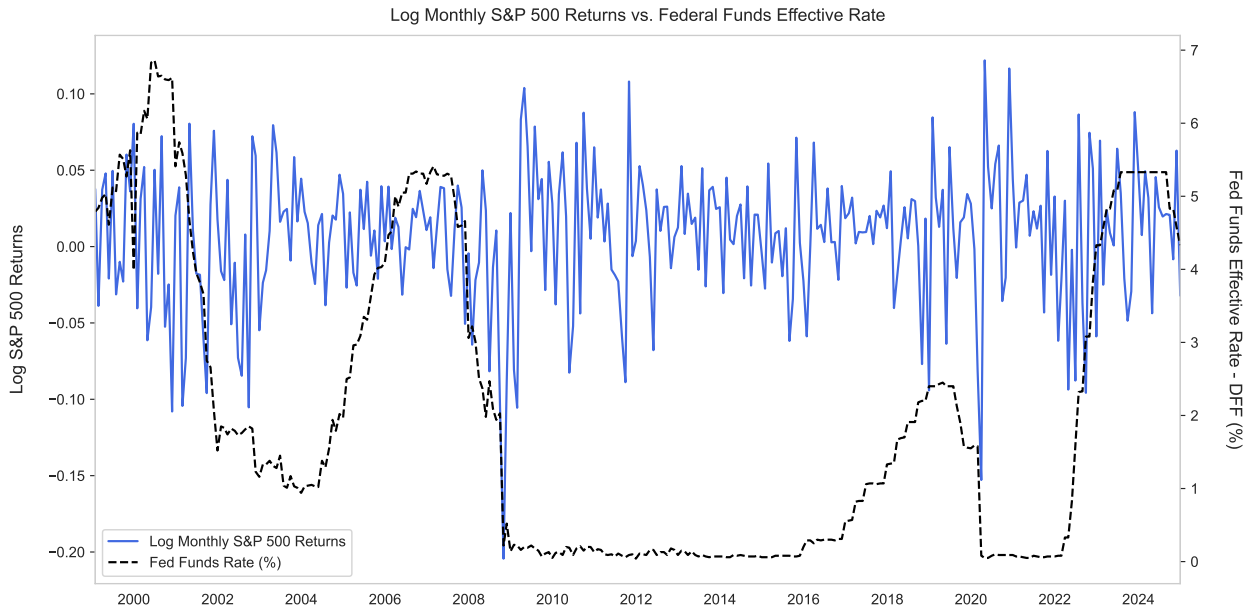


asset prices, with the shock of the policy decision dissipating over time.

Bernanke and Gertler (2000) also explore the impact of monetary policy on the volatility of asset prices, showing that changes in short-term interest rates significantly affect financial markets by altering investors' expectations and risk tolerance, core components of central asset pricing models. Specifically, their work suggests that *low interest rates increase risk-taking behavior* by reducing the returns on safer assets, thus encouraging investors to seek higher returns in riskier investments (e.g., increase a portfolio's weight on stocks). In contrast, *during periods of higher interest rates, risk tolerance tends to decrease* as the cost of borrowing rises, and the opportunity cost of holding risky assets becomes more apparent (e.g., increasing the portfolio weight on bonds). This behavior is consistent with findings in Christiano, Eichenbaum, and Evans (2005), who argue that tightening monetary policy and increasing rates lead to greater uncertainty, which discourages speculative investments such as stocks.

Whether during periods of low or high interest rates, monetary policy not only affects asset prices but also drives a broader re-evaluation of risk across financial markets. These

Figure 3: Log Equity Returns and the Federal Funds Effective Rate, 1999 to 2024



shifts in investors' risk tolerance are crucial for understanding how interest rate regimes influence investor behavior over time. Risk-taking in periods of market volatility across asset classes would reasonably differ according to how sensitive investors are to gains and losses, given such changes in asset prices and the evolving macroeconomic outlook. Moreover, the incentive structure changes regarding the demand for stocks and bonds as the Federal Reserve maintains a certain target range for the federal funds rate, in turn impacting risk-taking behavior during an interest rate regime.

Loss aversion measures the sensitivity to losses realtive to gains, with losses weighing on investors' value function greater than gains. During periods of low interest rates, rising asset prices and elevated expected return from risky securities would lead investors to underestimate risk. Therefore, hypothesized implied loss aversion is expected to be lower, giving way to portfolios weighted towards risky assets. Conversely, during periods of high interest rates (associated with economic tightening, uncertainty, and falling asset prices), investors would become more sensitive to losses. Therefore, hypothesized implied loss aversion is expected to be higher, with investors decreasing their appetite for risky investments and being

incentivized to pursue safer assets.

3 Increased Presence of Individual Investors and Their Behavior

Policy makers, asset managers, investment advisors, and individuals themselves must take into consideration how changes in monetary policy impact financial market activity and the effects of these changes on risk tolerance. This yields the central question of this paper: How would an individual investor's loss aversion change across different interest rate environments? Appetites for risk reasonably vary across individual investors with presumably heterogenous preferences, motivating an analysis of evaluating portfolio returns from the viewpoint of different investors.

Research shows that the pandemic acted as a catalyst for increased retail investor participation. In the U.S., for example, Wheat, O'Brien, and Eckerd (2024) reported that portfolios of retail investors became significantly more exposed to market risk from 2019 to 2021, with individuals transferring large sums from checking accounts into brokerage accounts during this period. This surge was partly driven by government stimulus payments, free time during the pandemic's lockdowns, and increased accessibility to commission-free trading platforms like Robinhood. In fact, Pagano, Sedunov, and Velthuis (2021) noted that platforms such as Robinhood saw record levels of activity during this time.

O. Akin and S. Akin (2024) reveal the profound impact of investor behavioral biases, loss aversion and herding behavior in particular, on stock market (S&P 500) performance. The authors show that as real interest rates rise, investors tend to become more risk-averse, leading to a shift toward safer assets, which in turn contributes to equity market declines. This negative relationship between interest rates and stock prices, as highlighted in their study, is further compounded by loss aversion. Investors, being more sensitive to potential losses than to gains, gravitate towards safe-haven assets and away from riskier assets.

In response to market volatility, investors also exhibit herding behavior and follow the actions of other market participants (for further discussion of herding behaviors of retail investors, see Hsieh, Chan, and Wang (2020)). O. Akin and S. Akin (2024) emphasize that during periods of high volatility, such as those induced by rising interest rates or other signals of uncertainty over future economic growth, investors tend to react more impulsively, ampli-

fying the negative impact of market fluctuations. The increased presence of retail investors during the pandemic, as discussed earlier, could exacerbate these behaviors, especially when consumer confidence shifts, further fueling market volatility.

This body of research underscores the growing importance of understanding individual investor behavior, particularly in periods of market turmoil. The increasing number of retail investors participating in the market, as seen during and after the COVID-19 pandemic, has significant implications for market stability. Given their heightened sensitivity to market sentiment and volatility—shaped by biases like loss aversion and herding behavior—retail investors are likely to contribute to the amplification of market fluctuations. Thus, analyzing how investors with different attitudes toward risk react to changing interest rate regimes is crucial for understanding broader market dynamics, both from the perspective of professional investors acting on behalf of their clients' best interests and from that of policymakers aiming to stabilize economic growth or act on behalf of their constituents.

3.1 Portfolio Allocation in a Prospect Theory Framework

Portfolio allocation is a critical decision-making process for individual investors, which becomes increasingly complex when considering the role of behavioral biases on investment choices. Models, such as the seminal mean-variance (MV) optimization of Markowitz (1952), offer a quantitative framework for maximizing returns for a given level of risk, or minimizing the variance of returns for a given level of expected return. However, these models assume rational behavior with optimizing agents, which does not align with observed investor behavior in financial markets. As such, alternative frameworks have emerged to provide a more accurate representation of how investors make decisions under risk and uncertainty.

MV optimization laid the foundation for modern portfolio theory by introducing the concept of diversification to minimize risk and enhance risk-adjusted returns. Under the framework in Markowitz (1952), a rational investor would construct an optimal portfolio by considering the expected return and risk-adjusted variance of returns, selecting assets that provide the highest expected return and lowest variance for a given level of risk tolerance. While the MV approach remains a central starting point in portfolio theory, it carries certain limitations in addressing investor biases. Elton and Gruber (1997) reviewed the MV

approach, noting its success in portfolio construction but also highlighting its inability to account for behavioral biases such as loss aversion and non-linear probability weighting, a caveat addressed by CPT's fundamental assumptions and use of decision weights.

CPT serves as a more accurate representation of how investors behave when faced with uncertainty, such as in spurts of market volatility or during periods of time wherein risk-taking behavior is influenced by external forces, such as the Federal Reserve's target range for the federal funds rate. Consigli, Hitaj, and Mastrogiacomo (2017) applied CPT to portfolio optimization, showing how an investor's risk preferences influence their optimal portfolio allocation. Their analysis reveals that investors who exhibit higher loss aversion tend to hold more diversified portfolios, as they are more sensitive to the potential for losses. Conversely, investors with a higher tolerance for risk may favor less diversified portfolios with higher expected returns. This makes the CPT framework particularly relevant when analyzing investor behavior during periods of high market volatility, such as the periods surrounding monetary policy implementations, and over the course of different interest rate regimes.

Incorporating behavioral finance into portfolio theory aims to provide a more realistic framework for understanding how individual investors make asset allocation decisions under risk and uncertainty. CPT is central to this as it accounts for loss aversion, probability distortion, and risk preferences in both gains and losses, offering potential insights into portfolio construction that are closer to reality. H. Levy and M. Levy (2004) found that the results of portfolio optimization under an MV approach nearly coincide with those of a prospect theory framework. Assuming normally distributed returns, they show that the MV-efficient frontier and the prospect-theory-efficient set are nearly identitcal. Hens and Mayer (2014) extended their analysis to the cumulative iteration of prospect theory, developing a novel computational strategy for computing optimal portfolios under CPT. Their analysis supports a distinct difference between MV and CPT portfolio optimization for non-normally distributed returns. These works serve as the basis for the methodology of this paper, aligning with Hens and Mayer (2014) and Consigli, Hitaj, and Mastrogiacomo (2017) in terms of the non-Gaussianity of the distribution of asset returns. Assuming a normal distribution of asset returns allows for smoother statistical computations but deviates from the "fat tails" observed in the data. As this paper aims to maintain a real-world perspective in the analysis of investor loss aversion over the course of different interest rate regimes, a non-normal

distribution is utilized to reflect the empirical behavior of asset returns.

4 Data

To mimic the investment strategies of real-world individual investors, data on the returns of two asset classes—stocks and bonds—are used to construct the portfolio of each investor type, referred to as risk-averse, risk-neutral, and risk-seeking. Equity holdings in simulated portfolios are represented by historical daily returns on the S&P 500 Index, inclusive of dividend distributions to capture total continuous compounding. Daily returns of the ICE Bank of America Corporate Bond Index (ICE Index) are used to represent the portfolio's second asset class. This index contains U.S. investment-grade corporate bonds and is widely referred to as a benchmark for fixed-income performance by the broader investment community. Returns on both indexes are collected from January 1, 1999, to December 31, 2024 from the Center for Research in Security Prices (CRSP), exclusive of null values due to non-trading days—generally market holidays. This sample of historical asset returns hones in on the many monetary policy implementations over the past two decades in response to several economy-wide events (financial market crashes, global pandemic, etc.). To assess the distribution of the data, Jarque-Bera Tests were performed on the log returns of the two series, which resulted in p-values of 0.0000 at the 0.05 significance level for both series. Normality is then rejected at the 0.05 significance level.

Generalizing the portfolio of the individual investor in this respect is unlikely to stray too far from reality. Retail investors largely hold diversified products representing baskets of stocks or bonds. Joshi and Dash (2024) note the rise of exchange-traded fund (ETF) activity by retail investors in recent years, and Easley et al. (2021) emphasize the use of ETFs as typical assets in a portfolio rather than a replication of its underlying securities; the latter would imply longer-term passive investing. While many may view stocks as the focus of traditional retail investment, holdings of bonds, particularly corporate bonds, by retail investors have increased substantially in recent years (see Haan, Li, and Watts (2023)). This motivates an analysis that replicates the payoffs of holding baskets of equities and corporate bonds. Additionally, risk-adjusted returns offer the notion that investors demand performance in excess of a risk-free rate. This can be modeled in the data by assuming the

yield on 3-month Treasury Bills as a risk-free rate. Data for the risk-free rate is obtained from the St. Louis Federal Reserve's Federal Reserve Economic Data (FRED).

5 Methodology

5.1 Overview of Dynamic Portfolio Optimization Under CPT

The portfolio optimization process under CPT is a more complex approach than mean-variance (MV) optimization due to the S-shape, piecewise nature of the CPT utility function. The seminal works of Andersen and Bollerslev (1997); Andersen and Bollerslev (1998) and Engle (2001) on general autoregressive heteroskedasticity (GARCH) models, particularly the time-varying form provided by Engle (2002), allow for estimating the conditional mean and volatility of time series data. Portfolios are then constructed using these conditional moments.

The core strategy implemented here is a nested optimization procedure, inspired by the convergence findings of H. Levy and M. Levy (2004). Instead of directly setting a loss aversion parameter λ to find the optimal CPT portfolio weight (w_{CPT}), the problem is reversed. First, a target optimal Mean-Variance weight (w_{MV}) corresponding to a known investor type (θ) is defined. Second, a search is conducted for the CPT loss aversion parameter ($\hat{\lambda}$) that minimizes the distance between the two weights. This process allows for the extraction of the time-varying, implied loss aversion parameter ($\hat{\lambda}$) of the three stylized investors under CPT, via different representative values of the MV risk-aversion parameter (θ). Constant parameters are assumed for probability weighting, concavity over gains, and convexity over losses, aligning with the empirical calibration of Tversky and Kahneman (1992).

The full simulation unfolds monthly:

1. Approximately one year of historical daily observations are used to estimate the conditional mean and covariance of the assets' excess returns using a DCC-GARCH model.

2. A target optimal stock weight (w_{MV}) is calculated for the next month using the standard MV framework, parameterized by the investor's risk aversion (θ).

3. The nested (inverse) optimization then finds the $\hat{\lambda}$ that forces the CPT optimal weight (w_{CPT}) to match this w_{MV} target.

4. This $\hat{\lambda}$ is recorded as the investor's perceived loss aversion for the month.

This methodology provides a robust, granular view of $\hat{\lambda}$ over time, ultimately leading to how investor behavior may shift during different economic and interest rate environments.

5.2 Models of Time-Varying Volatility

To capture the dynamic uncertainty of asset prices, the conditional variance of excess returns above the 3-month Treasury bill yield are employed, assuming this yield to be the risk-free rate for each month. A DCC-GARCH model is necessary to capture both the autoregressive nature of the conditional variance and the time-varying correlation between the two assets. Following Engle (2002), the specification for the conditional variance (H_t) is:

$$H_t = D_t^{1/2}\, R_t\, D_t^{1/2}, \quad D_t = \mathrm{diag}\big(h_{1,t},\, h_{2,t}\big), \tag{1}$$

$$h_{i,t} = \alpha_{0,i} + \alpha_{1,i}\, u_{i,t-1}^2 + \gamma_i\, h_{i,t-1} \quad (i = 1, 2), \tag{2}$$

$$R_t = \Big(\mathrm{diag}\, Q_t\Big)^{-1/2} Q_t \Big(\mathrm{diag}\, Q_t\Big)^{-1/2}, \tag{3}$$

$$Q_t = (1 - a - b)\, \overline{Q} \;+\; a v_{t-1}\, v_{t-1}^{\top} \;+\; b Q_{t-1}, \quad v_t = D_t^{-1/2}\, u_t, \tag{4}$$

where u_t are residuals of a GARCH(1,1) process implemented for each asset, $\overline{Q}$ is their unconditional covariance, and (a,b) govern shock- and persistence-effects in the conditional correlation. The output is the one-period-ahead conditional mean vector μ and covariance matrix $\Sigma \equiv H_t$. These are then scaled to represent a one-month forecast used by the investor to construct their portfolio, assuming 20 days in a month.

5.3 Mean-Variance Target Portfolio Construction

The conditional mean (μ) and covariance (Σ) forecasts from the DCC-GARCH model are first used to establish a benchmark portfolio w_{MV}. This target is the optimal weight that a standard MV investor, defined by their risk aversion θ, would choose.

The MV investor maximizes the following objective function:

$$\max_{w} \Big\{ w^{\top}\mu \;-\; \theta\, w^{\top}\Sigma\, w \Big\} \quad \text{s.t.} \quad \mathbf{1}^{\top} w = 1, \quad w \geq 0, \tag{5}$$

where w is the vector of portfolio weights, and θ is the risk-aversion parameter. Short-selling is restricted to focus on common retail investor constraints.

To stylize different investor types, θ is calibrated as follows:

- Risk-Averse: $\theta = 5.0$ (High Risk Aversion)
- Risk-Neutral: $\theta = 1.0$ (Minimal Risk Aversion)
- Risk-Seeking: $\theta = -5.0$ (High Risk Seeking)

5.4 Cumulative Prospect Theory (CPT) Framework

5.4.1 The Reference Point

Core to CPT is the reference point (ρ) from which returns are evaluated as gains or losses. In this analysis, the reference point is the risk-free rate, estimated using the 3-Month Treasury Bill. This conforms with the literature and provides an economically meaningful trade-off for the investors simulated in this analysis.

5.4.2 The CPT Value Function

Each investor's utility is represented by the piecewise value function derived from Tversky and Kahneman (1992). The CPT value function for an outcome x (portfolio return minus a reference point) is:

$$V(x;\alpha;\beta;\lambda) \;=\; \begin{cases} x^{\alpha}, & x \geq 0, \\ -\lambda\left(-x\right)^{\beta}, & x < 0, \end{cases} \tag{6}$$

with α and β controlling the curvature of the gain and loss domains, respectively, and λ capturing loss aversion. Parameters α and β are set as $\alpha = \beta = 0.88$, following Tversky and Kahneman (1992). This leaves λ as the free parameter to be extracted.

5.4.3 Decision Weights (π_i) and Probability Weighting

Decision Weights (π_i) The decision weight π_i assigned to each discrete outcome r_i is determined by the rank-dependent probability weighting (or weighting of the ranks) method

introduced in Tversky and Kahneman (1992). This transforms the cumulative distribution of the outcomes into a set of weights that overweigh small probabilities and underweigh intermediate/large probabilities.

The decision weight π_i is calculated by applying the probability weighting functions w^+ (for gains) and w^- (for losses) to the cumulative probability of the outcomes, P_i.

- **For Gains (Outcomes $r_i \geq \rho$):** The decision weight π_i^+ is derived from the difference in the cumulative probability weighting of the ranked outcomes.
- **For Losses (Outcomes $r_i < \rho$):** The decision weight π_i^- is derived from the difference in the cumulative probability weighting of the ranked outcomes, measured from the extreme loss.

The rank-dependent decision weights π_i are defined as:

$$\pi_i = \begin{cases} w^+(P_i) - w^+(P_{i+1}) & \text{for } r_i \geq \rho \text{ (Gains)} \\ w^-(P_i) - w^-(P_{i-1}) & \text{for } r_i < \rho \text{ (Losses)} \end{cases} \tag{7}$$

Where:

- P_i is the cumulative probability of all outcomes *at least as good as* r_i.
- $w^+(P)$ and $w^-(P)$ are the Tversky-Kahneman probability weighting functions:

$$w^+(P) = \frac{P^\gamma}{\left(P^\gamma + (1-P)^\gamma\right)^{1/\gamma}}, \quad w^-(P) = \frac{P^\delta}{\left(P^\delta + (1-P)^\delta\right)^{1/\delta}}.$$

The empirical Tversky-Kahneman parameters of $\gamma = 0.61$ and $\delta = 0.69$ are used for consistency with empirical findings.

5.5 Inner CPT Optimization (Maximizing Expected Utility)

The optimal CPT weight w_{CPT} is the weight that maximizes the CPT Expected Utility $E_{CPT}[V(R_p^e)]$. This involves a Riemann sum over N discrete outcomes r_i formed from a

fitted Student's t-distribution, chosen to conform to the non-Gaussianity of the assets' logged returns:

$$w_{CPT}(\lambda) = \arg\max_{w} E_{CPT}[V(R_p^e)] = \arg\max_{w} \left\{ \sum_{i=1}^{N} V(r_i - \rho) \cdot \pi_i \right\}$$

where ρ is the reference point (defined below) and π_i are the Tversky-Kahneman decision weights.

5.6 Nested (Inverse) Optimization

The central objective of this study is the extraction of the monthly implied loss aversion parameter $\hat{\lambda}$ that makes the CPT portfolio weight match the target w_{MV}.

Finally, the inverse optimization is implemented by finding the $\hat{\lambda}$ that minimizes the squared distance between the two optimal weights:

$$\hat{\lambda} \;=\; \arg\min_{\lambda} \Big(w_{CPT}(\lambda) - w_{MV}(\theta) \Big)^2 \quad \text{s.t.} \quad \lambda \in [1.0, 6.0] \tag{8}$$

The resulting time series of $\hat{\lambda}$ for each investor type constitutes the primary output of this study.

5.7 Interest Rate Regimes

To link behavioral shifts to macroeconomic factors, the investor's environment is categorized based on the 3-month Treasury Bill yield, calculated on a discount basis. For simplification, a quartile approach is used to distinguish three different interest rate regimes:

1. Low Rate (below 25th percentile, Q1)
2. Moderate Rate (above 25th, below 75th, Q2-Q3)
3. High Rate (above 75th percentile, Q4)

The median $\hat{\lambda}$ is then analyzed across these regimes to assess the impact of interest rates on implied loss aversion. While regime-switching literature offers a more robust definition of interest rate regimes, the simplification here allows for easier understanding of the proposed portfolio optimizing scheme as well as a straightforward analysis of loss aversion over time.

6 Results

Three investor types are simulated, risk-averse (RA, $\theta = 5.0$), risk-neutral (RN, $\theta = 1.0$), and risk-seeking (RS, $\theta = -5.0$), to extract their implied loss aversion ($\hat{\lambda}$) across different interest rate environments. One-month-ahead conditional mean and variance forecasts are generated for the S&P 500 and ICE Index returns, along with the time-varying correlation, using the DCC-GARCH model. The full covariance matrix is constructed to solve the mean-variance objective function, which provides the target optimal portfolio weight (w_{MV}) for each investor. The monthly CPT-implied loss-aversion parameter $\hat{\lambda}$ is then backed out by minimizing the squared distance between the CPT weight (w_{CPT}) and the MV weight (w_{MV}). Each month is classified into one of the defined regimes based on the monthly yield on the 3-Month Treasury Bill (3MTB).

The regime analysis, utilizing the quartile distribution of the 3-Month T-Bill rate to delineate Low ($\leq$ Q1), Normal (Q1 to Q3), and High ($\geq$ Q3) regimes, yields strong evidence for the time-varying nature of investor behavior. The central finding is that implied loss aversion ($\hat{\lambda}$) systematically changes across interest rate regimes, and these changes are statistically significant, particularly for the Risk-Averse investor.

Table 2 reports the median estimated $\hat{\lambda}$ for each investor per quartile regime, along with their standard deviations and the results of the Mann-Whitney U tests comparing the extreme regimes to the moderate one. Table 1 displays the computed T-Bill rate thresholds defining these regimes.

The use of the nested (inverse) CPT method successfully provides an estimate for $\hat{\lambda}$ for all months in the chosen timeframe. Bounds of 1.0 and 6.0 were placed on the loss aversion parameter in the event of non-finite values derived from the piece-wise function. Initializations of the loss aversion parameter were 1.0, 2.25, and 5.0. The value function creates complexity when returns are entirely one-sided (i.e., a month's returns are only gains and zero losses), necessitating a grid search fallback when the numerical optimization fails.

Table 1: 3MTB Rate Quartile Thresholds (Annual Percentage Points)

Regime	25th (Q1)	50th (Q2)	75th (Q3)	Max
Rate (%)	0.11	1.18	3.37	6.19

Table 2: Implied Loss Aversion (λ) Estimates Across Interest Rate Regimes

Investor Profile	**Low Rates** ($\leq$ Q1)	**Mod Rates** (Q1-Q3)	**High Rates** ($\geq$ Q3)	**MWU** p**-val** (Low-Mod)	**MWU** p**-val** (High-Mod)
Risk-Averse ($\theta = 5.0$)					
Median $\hat{\lambda}$	**1.52**	1.71	2.25	0.00	0.38
Std. Dev.	0.46	0.37	0.39		
Sample Size (N)	55	103	53		
Risk-Neutral ($\theta = 1.0$)					
Median $\hat{\lambda}$	**1.00**	1.01	1.01	0.00	0.33
Risk-Seeking ($\theta = -5.0$)					
Median $\hat{\lambda}$	**2.71**	2.59	2.50	0.00	0.55

Note: This table summarizes the distribution of the monthly implied loss aversion parameter.

7 Discussion

The simulation yields two key findings regarding the median implied loss aversion ($\hat{\lambda}$). First, the primary result is the systematic, statistically significant directional movement of implied loss aversion relative to the prevailing interest rate regime. This directly validates the core hypothesis of this study. For the Risk-Averse investor, the median implied loss aversion is found to be lowest in the Low-Rate regime and highest in the High-Rate regime. Specifically, the Low-Rate regime (below Q1) exhibits a statistically different median $\hat{\lambda}$ from the Moderate regime, suggesting that when the risk-free rate is historically low, investors become less loss averse and are encouraged to allocate more heavily toward risky assets. Conversely, the High-Rate regime (above Q3) shows a median $\hat{\lambda}$ that is economically significant due to its high value relative to the median in the Moderate regime. This suggests that during periods of high rates, loss aversion rises, leading to a psychological shift toward conservative portfolio allocation. This finding strongly reinforces the view that the market's psychological state systematically responds to macroeconomic factors.

Second, the $\hat{\lambda}$ values display a clear and arguably counterintuitive rank-ordering by investor type. The Risk-Seeking (RS, $\theta = -5.0$) investor exhibits the highest loss aversion ($\hat{\lambda}$), while the Risk-Averse (RA, $\theta = 5.0$) investor shows a lower loss aversion ($\hat{\lambda}$). The Risk-Neutral (RN, $\theta = 0.0$) investor remains loss-neutral ($\hat{\lambda} = 1.0000$).

- Risk-Neutral Investor: $\hat{\lambda} = 1.0000$ is a logical result, as a loss-neutral CPT investor is equivalent to a simple expected value maximizer, which often aligns closely with the risk-neutral MV target ($\theta = 0$).

- Risk-Seeking Investor: The high $\hat{\lambda}$ (typically ≈ 5) suggests that to force the CPT portfolio to match the highly aggressive MV target, the CPT investor must be endowed with an extremely high penalty for losses. This reflects the structural necessity of the inverse optimization.

- Risk-Averse Investor: The $\hat{\lambda} \approx 2.25$ is close to the canonical value of 2.25 found by Tversky and Kahneman (1992). Its magnitude relative to the RS investor is expected because the RA investor's MV target is a low-risk, low-stock portfolio, which requires a lower $\hat{\lambda}$ to be matched by the loss-averse CPT framework.

This implementation provides solid grounds from which to expand the analysis and include more rigorous, time-varying interest rate regimes, investor risk parameters, and measures of loss aversion from a CPT prospective utility function. Further, alternative methods of computing the latent loss aversion parameter can be considered, as well as alternative formulations of the CPT value and probability weighting functions.

8 Conclusion

This paper integrates time-varying conditional volatility of returns on the S&P 500 Index and ICE Corporate Bond Index to simulate portfolios under a CPT framework and extract the loss aversion (λ) of three investor types. By extracting an implied λ from minimizing the distance between a CPT-maximizing and mean-variance-maximizing investor, a time series of loss aversion parameters is estimated. This allows for the study of how investors adapt their portfolio allocations and exhibit changing loss aversion across different environments, with this paper focusing on distinct monetary policy regimes. The simulations yielded the central finding that the implied loss aversion parameter ($\hat{\lambda}$) is systematically dependent on the interest rate environment. Specifically, $\hat{\lambda}$ decreases in low-rate regimes, encouraging a higher allocation to risky assets, and increases in high-rate regimes, driving investors toward safer assets. The Mann-Whitney U test results confirm these shifts are statistically significant for the Risk-Averse investor when comparing low regimes to a moderate one, while economically significant when comparing high regims to a moderate one. These results reinforce the view that the market's psychological state responds systematically to monetary policy and macroeconomic uncertainty.

These findings highlight the importance of combining behavioral models with volatility dynamics for understanding real-world allocation decisions. For practitioners, tracking implied loss aversion over time may serve as an early warning indicator of shifting risk preferences in retail flows. For individuals themselves, increasing one's awareness of biases may enable stronger portfolio strategies to avoid excessive losses during periods of market turbulence. Future research could extend this framework to multi-asset settings, consider alternative risk-profile assumptions, incorporate regime-specific CPT parameters, or explore the predictive power of loss aversion for market returns and flows.

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